

Bruna Caroline Campos

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Professional Summary

Civil engineer with over eight years of experience in computational mechanics and finite element modeling for structural, geomechanical, and multiphysics systems. Skilled in C++, Python, and open-source simulation software (deal.II, Lethe), CAD preparation, mesh generation, solver development, and high-performance computing. Strong background in numerical modeling, verification and validation, and engineering design analysis. Experienced in interdisciplinary collaboration and technical communication.

Technical Skills

Languages: English (fluent), French (intermediate), Portuguese (native)

Programming: C++, Java, MATLAB, Python, Bash Script

Simulation Software: Lethe, Ansys, Revit, TQS, Abaqus

CAD and Meshing: Onshape, Fidelity Pointwise, SolidWorks, AutoCAD

Open-source Development: Lethe (CFD), deal.II (finite element libraries), INSANE (structural analysis)

High-Performance Computing: MPI/OpenMP, matrix-free algorithms

Development Environments & Tools: Eclipse, VS Code, Sublime Text, Git

Numerical Methods: Computational Mechanics, Structural Analysis, Porous Media Modeling, Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), Multiphysics Simulation

Professional and Research Experience

Postdoctoral Fellow — Chemical Engineering, Polytechnique Montréal 2025–Present

- Develop open-source CFD FEM models in *Lethe* for simulating rotating reactors
- Couple domain decomposition methods in C++ with high-performance computing
- Collaborate with experimental teams to validate and optimize simulations of industrial processes

PhD Researcher — Civil and Environmental Engineering, University of Waterloo 2021–2024

- Designed and implemented finite element formulations for coupled porous media systems
- Conducted analyses for various length scales, source frequency ranges, and porous material types
- Published three peer-reviewed journal papers on multiphysics modeling
- Applied numerical modeling to engineering problems in geomechanics and energy storage

Teaching and Mentoring Roles 2017–2024

- Teaching Assistant, Finite Element Analysis – University of Waterloo (2022–2024)
- English Instructor – Red Balloon School and Private Lessons (2020–2022)
- Teaching Assistant, French Language – Federal University of Minas Gerais (2019)
- Teaching Assistant, Structural Analysis II – Federal University of Minas Gerais (2018)
- Teaching Assistant, Structural Analysis II – Federal University of Uberlândia (2017)

Project Manager Intern — Conel Construction Inc., Brazil 2016–2017

- Collaborated in structural design activities and construction scheduling for building projects
- Participated in budget planning and elaboration of contracts

Selected Projects and Publications

REDEVELOP Challenge: Compressed Air Energy Storage – University of Calgary 2021–2022

- NSERC-funded project for Collaborative Research and Training Experience
- *Compressed Air Energy Storage in salt caverns: a solution to renewable energy intermittency in Alberta?*
- In-depth literature review of energy storage techniques, establishing a connection between interests of industry and indigenous communities

Recent Publications

- Campos, B., Munch, P., Ferreira, V. O., Boffito, D. C., Banquy, X., Blais, B. (2026), High-order stabilized matrix-free simulation of rotating mixing devices using the mortar element method. arxiv.org/abs/2608.27423
- Campos, B., Munch, P., Ferreira, V. O., Blais, B. (2026) On the simulation of rotating devices using the mortar element method on a coupled continuous-discontinuous Galerkin formulation. 17th World Congress on Computational Mechanics.
- Campos, B., Munch, P., Ferreira, V. O., Blais, B. (2026), Numerical simulation of rotor-stator devices using a sliding-mesh matrix-free computational fluid dynamics model. 4th International Process Intensification Conference (IPIC 4).
- Campos, B., Gracie, R. (2025), Multiphysics Wave Simulator for Generalized Porous Media Across Scales. *Advances in Engineering Software*, 218. doi:10.1016/j.advengsoft.2026.104172
- Campos, B., Gracie, R. (2024), A dynamic three-field finite element model for wave propagation in linear elastic porous media. *Int. J. Numer. Anal. Methods Geomech.*, 49(4). doi:10.1002/nag.3916
- Campos, B., Gracie, R. (2023), A FEM for three-field u-p- η poroelasticity with nonreciprocal interactions, *Int. J. Numer. Anal. Methods Geomech.*, 48(1). doi:10.1002/nag.3636

Education

PhD, Civil and Environmental Engineering – University of Waterloo, Canada 2021–2024

Thesis: Finite Element Models for Multiscale Porous Media

MASc, Structural Engineering – Federal University of Minas Gerais (UFMG), Brazil 2018–2020

Thesis: Numerical Integration by Subdivision of Elements using the GFEM

BASc, Civil Engineering – Federal University of Uberlândia (UFU), Brazil 2012–2017

Thesis: Structural solutions for concrete slabs with high conduit concentration

Awards and Distinctions

- G.E. Oravas Memorial Scholarship, McMaster University (2024)
- Activity Report Award, University of Waterloo (2024)
- REDEVELOP Scholarship and REDEVELOP Challenge Winner (2022)
- Brazilian Math Olympiad, Gold Medal (2009) and Bronze Medal (2007)

Volunteering

- Professional Development Director, Women in Engineering (WiE) – University of Waterloo (2023–2024)
- Math Tutor – United for Literacy, Canada (2023)
- English Teacher and Coordinator – Program *Incluir*, Brazil (2019–2022)

Additional Information

- Open-source software contributor: Lethe (CFD-DEM) and INSANE (Structural Analysis)
- Experienced in technical communication, project coordination, and interdisciplinary collaboration
- Eligible to work in Canada